

A Direct Record-Increment Description of the Bandlow–Killpatrick Bijection

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July 2026

1 Abstract

We present a direct record-increment description of the inverse of the Bandlow–Killpatrick bijection between 312-avoiding permutations and Dyck paths. The construction replaces the original adjacent-transposition algorithm by a direct encoding based on successive increases of left-to-right maxima. The associated Dyck path is determined entirely by its sequence of record increments. We prove that this construction is a bijection. Then we show that it coincides exactly with the inverse of the Bandlow–Killpatrick map.

As a consequence, the map acquires the lattice isomorphism between the right weak order on 312-avoiding permutations and the Tamari lattice on Dyck paths. Beyond this, the record-increment formulation yields simple and explicit formulas for several natural statistics. In particular, records correspond to peaks.

2 Introduction

Pattern-avoiding permutations and Dyck paths are among the most studied Catalan families in combinatorics. Numerous bijections explain their common enumeration by the Catalan numbers.

Among the many bijections connecting Catalan families, several explicit correspondences have been established between 312-avoiding permutations and Dyck paths. Independently, Björner and Wachs defined a map ($\beta_{\min}$) that associates 312-avoiding permutations with binary trees. More recently, Bandlow and Killpatrick constructed an area-to-inversion bijection. This map has order-preserving properties and realizes a lattice isomorphism between the right weak order on 312-avoiding permutations and the Tamari lattice on Dyck paths.

In this paper, we provide a record-increment description of the inverse of the Bandlow–Killpatrick bijection from Dyck paths to 312-avoiding permutations. Unlike the original construction, which is defined through a sequence of adjacent transpositions, our map is obtained from a single left-to-right scan of the permutation.

The Dyck path can be reconstructed directly from the record increments. It gives a simple linear-time algorithm and provides direct access to several natural statistics transported by the bijection.

The main results of this paper are the following:

First, we introduce a new direct description of the inverse of the Bandlow–Killpatrick bijection. Our construction is formulated in terms of the record-increments of a 312-avoiding permutation. It does not rely on the sequence of adjacent transpositions that appears in the original definition.

Second, we show that this description naturally reflects the order structures involved. In particular, Consequently, the map inherits the lattice isomorphism between the right weak order on 312-avoiding permutations and the Tamari lattice on Dyck paths.

Third, we study statistics transported through the bijection. We prove, in particular, that record statistics correspond to peaks of Dyck paths and derive explicit formulas for statistics such as the height and our definition of area of Dyck paths.

Finally, we prove that this record-increment construction is indeed the inverse of the Bandlow-Killpatrick bijection. This establishes a new explicit realization of the correspondence between 312-avoiding permutations and Dyck paths.

To situate our contribution within this landscape, we summarize in the following table the main known bijections between 312-avoiding permutations and Dyck paths. Our bijection coincides with the inverse of the Bandlow-Killpatrick bijection. Therefore, the last two rows of the table describe the same correspondence from two different perspectives.

Author	Domain	Co-domain	Key invariant	Preserves Tamari lattice	Complexity
Bandlow-Killpatrick [BK01]	312	Dyck path	area statistics $\longleftrightarrow$ inv	Yes	$O(n)$
Present work	312	Dyck path	record increments (climb length)	Yes	$O(n)$

The Bandlow-Killpatrick bijection preserves the Tamari order. This follows from its description in terms of chains of adjacent transpositions. Since our bijection ϕ is a direct, self-contained, one-pass reformulation of its inverse (Section 8), it also preserves the Tamari order.

Although the Bandlow-Killpatrick bijection is algorithmically simple, its description is intrinsically dynamic, relying on successive adjacent transpositions. Our formulation replaces this evolving process by a static encoding extracted directly from the permutation.

3 Preliminaries

3.1 Dyck words and binary trees

Definition 3.1 (Dyck word). *A Dyck word is a word $w \in \{u, d\}^*$ satisfying:*

- (D1) $|u(w)| = |d(w)|$ (the number of u 's equals the number of d 's)
- (D2) for every prefix w' of w , one has $|u(w')| \geq |d(w')|$

We denote by $\mathcal{D}_n$ the set of Dyck words of length $2n$. Its cardinality is the n -th Catalan number C_n .

Definition 3.2 (Height). *For a word $w \in \{u, d\}^*$, its height is defined by: $h(w) = |u(w)| - |d(w)|$. A Dyck word is precisely a word w such that $h(w) = 0$ and $h(w') \geq 0$ for every prefix w' of w .*

Lemma 3.1 (Canonical decomposition). *Let $w \in \mathcal{D}_n$ be a non-empty Dyck word, which necessarily begins with u . Then there exists a unique pair (A, B) of Dyck words (possibly empty) such that: $w = u \cdot A \cdot d \cdot B$ where uAd is the shortest prefix of w that returns to height 0. This decomposition is unique.*

Proof. Since w is non-empty and must begin with u (indeed, $h(\emptyset) = 0$ and condition (D2) prohibits w from starting with d), we may write $w = u \cdot w'$.

Let k be the smallest index such that the prefix of w of length $k+1$ returns to height 0. Such an index exists because $h(w) = 0$.

We then set $A = w'[1 \dots k-1]$ and $B = w[k+2 \dots 2n]$ so that A consists of the letters of w between the positions 2 and k .

By minimality of k , the prefix uAd is the shortest prefix of w that returns to height 0, and therefore the decomposition $w = uAdB$ is unique. $\square$

Definition 3.3 (γ map). We define recursively the map $\gamma : \mathcal{D}_n \rightarrow \mathcal{T}_n$ by:

- $\gamma(\emptyset) = \text{empty tree (denote by } \emptyset\text{)}.$
- $\gamma(w) = \text{Node}(\gamma(A), \gamma(B))$ where $w = uAdB$ is the canonical decomposition of w .

Fact 3.1 (Huang-Tamari). The Tamari order on binary trees is defined by $A \leq B$ if A can be obtained from B by a sequence of right rotations.

The Tamari order on Dyck paths is defined by $A \leq B$ if A can be obtained from B by a sequence of local rotations shifting a rise to the right.

The map $\gamma : \mathcal{D}_n \rightarrow \mathcal{T}_n$ is a lattice isomorphism between the Tamari order on $\mathcal{D}_n$ and the Tamari order on binary trees $\mathcal{T}_n$.

Example 3.1. Let $w = uAdB$ and $A = ud, B = ud$. Then $\gamma(w) = \text{Node}(\gamma(A), \gamma(B))$

With: $A = ud, B = ud$

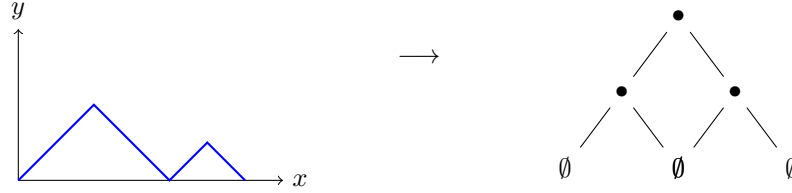
We first compute $\gamma(ud)$:

- the word $ud = u \cdot d$ has itself as the shortest prefix returning to height 0.
- hence its canonical decomposition is $ud = u \cdot \emptyset \cdot d \cdot \emptyset$.
- By definition, $\gamma(ud) = \text{Node}(\gamma(\emptyset), \gamma(\emptyset)) = \text{Node}(\emptyset, \emptyset)$, that is, a node whose two subtrees are empty.

We may now compute $\gamma(w)$:

Since $A = B = ud$, we obtain $\gamma(w) = \text{Node}(\gamma(A), \gamma(B)) = \text{Node}(\gamma(ud), \gamma(ud))$. So $\gamma(w) = \text{Node}(\text{Node}(\emptyset, \emptyset), \text{Node}(\emptyset, \emptyset))$

This yields the following tree:



3.2 312-avoiding permutations

Definition 3.4 (Containing a pattern). A permutation σ of length n is said to contain a pattern μ of length $k \leq n$ if there exists a subsequence w of σ with $|w| = k$ such that the relative order of the values in w is the same as that of μ .

Definition 3.5 (312-avoiding permutation). A permutation σ of length n is 312-avoiding if it does not contain the pattern 312.

Equivalently, there do not exist indices $i_1 < i_2 < i_3$ such that: $\sigma_{i_2} < \sigma_{i_3} < \sigma_{i_1}$

We denote by $Av_n(312)$ the set of permutations in S_n avoiding the pattern 312. We have $|Av_n(312)| = C_n$.

Definition 3.6 ($\beta_{\min}$ map). Let $\pi \in S_n, (n \geq 1)$. Let $m = \min(\pi)$ and let k be its position in π . We write the unique decomposition $\pi = L \cdot m \cdot R$ where $L = (\pi_1, \dots, \pi_{k-1})$ and $R = (\pi_{k+1}, \dots, \pi_n)$.

We define the map recursively by: $\beta_{\min}(\pi) = \text{Node}(\beta_{\min}(L), \beta_{\min}(R))$ with $\beta_{\min}(\emptyset) = \text{the empty tree}$.

Definition 3.7 (Right weak order). We define the right weak order $\leq_W$ in S_n by $\sigma \leq_W \pi \iff \text{Inv}(\sigma) \subseteq \text{Inv}(\pi)$

where $\text{Inv}(\pi) = \{(i, j) : i < j, \pi_i > \pi_j\}$ is the set of inversions of π .

Fact 3.2 (Björner & Wachs, 1991). *For every $\pi \in Av_n(312)$, the map $\beta_{\min}$ is a lattice isomorphism between $(Av_n(312), \leq)$ endowed with the right weak order and the poset of binary trees $\mathcal{T}_n$.*

Definition 3.8 (Tamari order on $Av_n(312)$). *We endow $Av_n(312)$ with the order induced by the right weak order $\leq_W$: $\pi \leq_W \sigma \iff \text{Inv}(\pi) \subseteq \text{Inv}(\sigma)$ ($\pi, \sigma \in Av_n(312)$)*
Björner-Wachs showed that $\beta_{\min}$ transports the right weak order restricted to $Av_n(312)$ to the Tamari order on binary trees.

Example 3.2. *Let $\pi = 213 \in S_3$*

Step 1: Identify the minimum and decompose. *The minimum of π is $m = 1$, appearing at position $k = 2$.*

Thus $\pi = (2) \cdot 1 \cdot (3)$.

So, $L = (2), R = (3)$.

By definition: $\beta_{\min}(213) = \text{Node}(\beta_{\min}(2), \beta_{\min}(3))$.

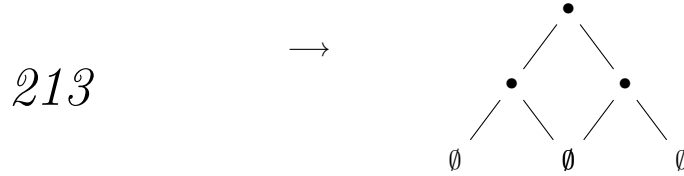
Step 2: Compute $\beta_{\min}(2)$. *The permutation (2) has minimum at position 1.*

Its decomposition is $\emptyset \cdot (2) \cdot \emptyset$.

Thus $\beta_{\min}(2) = \text{Node}(\beta_{\min}(\emptyset), \beta_{\min}(\emptyset))$, a leaf.

Step 3: Compute $\beta_{\min}(3)$. *Similarly, $\beta_{\min}(3) = \text{Node}(\emptyset, \emptyset)$.*

Step 4: Final assembly. $\beta_{\min}(213) = \text{Node}(\text{Node}(\emptyset, \emptyset), \text{Node}(\emptyset, \emptyset))$.



3.3 Record increase

While $\beta_{\min}$ provides a classical bijection from 312-avoiding permutations to binary trees, our goal is to construct a direct bijection to Dyck paths. To do so, we introduce the record-increase encoding.

Definition 3.9 (Record increase). *Let $L = (l_1, \dots, l_p)$ be a finite sequence of positive, distinct integers not necessarily consecutive. For every $i \geq 1$, define: $M_i(L) = \max(l_1, \dots, l_i)$, $M_0(L) = 0$. The record increase of L is the sequence $(\mu_1(L), \dots, \mu_p(L))$ defined by:*

$$\mu_i(L) = \begin{cases} l_i - M_{i-1}(L) & \text{if } l_i > M_{i-1}(L) \\ 0 & \text{else.} \end{cases}$$

When $L = \pi \in S_n$, we simply write $\mu_i = \mu_i(\pi)$ and $M_i = M_i(\pi)$.

Definition 3.10 (ϕ). *We define: $\phi(L) = u^{\mu_1(L)} d u^{\mu_2(L)} d \dots u^{\mu_p(L)} d$*

This definition applies to any sequence of positive distinct integers, and in particular to permutations.

Lemma 3.2 (Telescoping). *For every permutation $\pi \in S_n$ and every $k \in \{1, \dots, n\}$: $M_k = \sum_{i=1}^k \mu_i$*

Proof. We proceed by induction on k .

Base case: $k = 1$. π_1 is always a record because $M_0 = 0 < \pi_1$.

Thus, $\mu_1 = \pi_1 - 0 = \pi_1 = M_1$.

Induction step: Assume $M_{k-1} = \sum_{i=1}^{k-1} \mu_i$. We distinguish two cases:

- If $\pi_k > M_{k-1}$: then $M_k = \pi_k$ and $\mu_k = \pi_k - M_{k-1}$. So: $\sum_{i=1}^k \mu_i = M_{k-1} + (\pi_k - M_{k-1}) = \pi_k = M_k$
- If $\pi_k \leq M_{k-1}$: then $M_k = M_{k-1}$ and $\mu_k = 0$. Hence: $\sum_{i=1}^k \mu_i = M_{k-1} + 0 = M_{k-1} = M_k$ $\square$

Proposition 3.1 (Properties of μ). *Let $\mathcal{R}_n$ denote the set of sequences satisfying (C1)-(C3). For every permutation $\pi \in S_n$, the sequence $(\mu_1, \dots, \mu_n)$ belongs to $\mathcal{R}_n$, ie:*

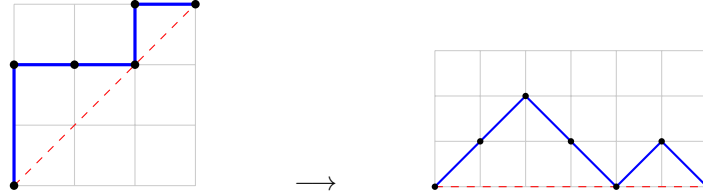
- (C1) $\mu_i \geq 0$ for all i .
- (C2) $\sum_{i=1}^n \mu_i = n$.
- (C3) for all $k \in \{1, \dots, n\}$, $\sum_{i=1}^k \mu_i \geq k$

Proof. (C1) By definition, $\mu_i = \pi_i - M_{i-1} > 0$ at record positions, and $\mu_i = 0$ otherwise.

(C2) By the telescoping Lemma 3.2, $\sum_{i=1}^n \mu_i = M_n = \max(\pi_1, \dots, \pi_n) = n$. Since π is a permutation of $\{1, \dots, n\}$.

(C3) Again by the telescoping lemma, $\sum_{i=1}^k \mu_i = M_k = \max(\pi_1, \dots, \pi_k)$. The k values $\pi_1, \dots, \pi_k$ are the k distinct integers from $\{1, \dots, n\}$, so their maximum is at least k . Thus, $M_k \geq k$. $\square$

Example 3.3. *In the left diagram, the vertical steps represent the successive values of the sequence L . The dashed diagonal shows the current maximum M_i . The right diagram shows the Dyck path obtained by applying ϕ , where each rise corresponds to a record increase.*



Proposition 3.2. *For every permutation $\pi \in S_n$, $\phi(\pi)$ is a Dyck word.*

Proof. The total number of u -steps is $\sum_{i=1}^n \mu_i = n$ by (C2) of Proposition 3.1, and the total number of d -steps is also n . Hence $h(\phi(\pi)) = 0$, which shows (D1).

For (D2), we check that the height remains positive after each block $u^{\mu_i}d$. After the first k blocks, the height is: $\underbrace{+1 + 1 \dots + 1}_{\mu_i \text{ times}} \underbrace{-1}_d$ the height increases by μ_i and decreases by 1.

Let $\sum_{i=1}^k (\mu_i - 1) = \left(\sum_{i=1}^k \mu_i \right) - k \geq 0$ by (C3) (because $\sum_{i=1}^k \mu_i \geq k$). Additionally, inside the block $u^{\mu_i}d$, the height increases by μ_i before it decreases by 1, so it stays positive all the time. Thus, $\phi(\pi)$ satisfies (D2). $\square$

4 Bijection ϕ

4.1 Recursive decomposition and standardization

Definition 4.1 (Standardization). *Let $L = (l_1, \dots, l_p)$ be a sequence of distinct integers. Standardization $\text{std}(L)$ is the unique permutation of $\{1, \dots, p\}$ such that: $\text{std}(L)_i < \text{std}(L)_j \iff l_i < l_j$. Standardization keeps the relative order of the elements, and in particular, pattern avoidance: if $\pi \in \text{Av}_n(312)$ and L is a subsequence of π , then $\text{std}(L) \in \text{Av}_{|L|}(312)$.*

Lemma 4.1 (Values distribution). *Let $\pi \in Av_n(312)$. After renormalizing so that the minimum is 1, we write $\pi = L \cdot 1 \cdot R$ with $p = |L|$. So: $Val(L) = \{2, \dots, p+1\}$, $Val(R) = \{p+2, \dots, n\}$*

Proof. By construction, $Val(L)$ and $Val(R)$ are disjoint and $Val(L) \cup Val(R) = \{2, \dots, n\} \setminus \{1\}$. The values of π other than 1 are divided between L and R .

First point: $Val(L) \subseteq \{2, \dots, p+1\}$

We argue by contradiction that there exists $i_1 \in \{1, \dots, p\}$, such that $\pi_{i_1} \geq p+2$. Let $l = \pi_{i_1}$.

Because $l \geq p+2$, the interval $\{2, \dots, l-1\}$ includes at least p elements. But $Val(L) \setminus \{l\}$ is a set of $p-1$ distinct elements in $\{2, \dots, n\} \setminus \{l\}$, so $Val(L) \setminus \{l\}$ can only hold $p-1$ elements in $\{2, \dots, l-1\}$. There remains at least one element of $\{2, \dots, l-1\}$ that is not included in $Val(L)$, and therefore is included in $Val(R)$.

Let $r \in Val(R)$ with $1 < r < l$, and let $i_3 > p+1$, be the position of r in π . Let $i_2 = p+1$, the position of the value 1 in π . We have $i_1 < i_2 < i_3$ and $\pi_{i_2} = 1 < \pi_{i_3} = r < \pi_{i_1} = l$

The subsequence $(\pi_{i_1}, \pi_{i_2}, \pi_{i_3})$ satisfies $\pi_{i_2} < \pi_{i_3} < \pi_{i_1}$, which constitutes a 312 pattern. This contradicts $\pi \in Av_n(312)$.

So $Val(L) \subseteq \{2, \dots, p+1\}$.

Second point: $Val(L) = \{2, \dots, p+1\}$

The set $\{2, \dots, p+1\}$ contains exactly p elements, and $Val(L)$ is a subset of $\{2, \dots, p+1\}$ of cardinality p . So $Val(L) = \{2, \dots, p+1\}$.

Third point: By complementarity, $Val(R) = \{2, \dots, n\} \setminus \{2, \dots, p+1\} = \{p+2, \dots, n\}$. $\square$

Corollary 4.1. *According to the hypothesis of Lemma 4.1, for all $i \in \{1, \dots, p\}$ and $j \in \{1, \dots, q\}$ with $q = |R|$: $L_i = \text{std}(L)_i + 1$, $R_j = \text{std}(R)_j + (p+1)$*

Proof. $Val(L) = \{2, \dots, p+1\}$ is a shift of $\{1, \dots, p\}$: standardization that subtracts 1 from each element. $Val(R) = \{p+2, \dots, n\}$ is a shift of $\{1, \dots, q\}$: the standardization subtracts $p+1$ from each element. $\square$

Lemma 4.2 (Recursive decomposition of $\phi(\pi)$). *Let $\pi \in Av_n(312)$ with $\pi = L \cdot 1 \cdot R$ (after the normalization for the minimum to be 1), $p = |L|, q = |R|$. Thus: $\phi(\pi) = u \cdot \phi(\text{std}(L)) \cdot d \cdot \phi(\text{std}(R))$*

Proof. Denote $(\alpha_1, \dots, \alpha_p)$ the record increases of the standardized permutation $\text{std}(L)$ and $(\beta_1, \dots, \beta_q)$ those of $\text{std}(R)$. We denote $m_i = M_i(\text{std}(L))$ and $b_j = M_j(\text{std}(R))$ the respective current maxima.

By definition of ϕ and by concatenation of $\pi = L \cdot 1 \cdot R$: $\phi(\pi) = \phi(L) \cdot u^{\mu_{p+1}(\pi)} d \cdot \phi(R)$

We compute each of the three terms.

Computation of $\phi(L)$ (case $p \geq 1$)

(If $p = 0$, then $L = \emptyset$, $\text{std}(L) = \emptyset$, and $\phi(L) = \phi(\text{std}(L)) = \varepsilon$ trivially. The computation below concerns the case $p \geq 1$, where L is non-empty.)

By Corollary 4.1, $L_i = \text{std}(L)_i + 1$ for all i , so $M_i(L) = m_i + 1$ for all $i \geq 1$ and $M_0(L) = 0$.

We distinguish two cases according to i :

- $i = 1$: L_i is always a record because $M_0(L) = 0 < L_1$. We have: $\mu_1(L) = L_1 - 0 = \text{std}(L)_1 + 1 = \alpha_1 + 1$

- $i \geq 2$: L_i is a record of L if and only if $L_i > M_{i-1}(L)$, this is equivalent to $\text{std}(L)_i + 1 > m_{i-1} + 1$, i.e. $\text{std}(L)_i > m_{i-1}$, i.e. $\text{std}(L)_i$ is a record of $\text{std}(L)$. Thus, for $i \geq 2$, L and $\text{std}(L)$ have exactly the same record positions. At record positions: $\mu_i(L) = L_i - M_{i-1}(L) = (\text{std}(L)_i + 1) - (m_{i-1} + 1) = \text{std}(L)_i - m_{i-1} = \alpha_i$

At non-record positions: $\mu_i(L) = 0 = \alpha_i$.

So we obtain: $\phi(L) = u^{\alpha_1+1}d \cdot u^{\alpha_2}d \cdots u^{\alpha_p}d = u \cdot (u^{\alpha_1}d \cdot u^{\alpha_2}d \cdots u^{\alpha_p}d) = u \cdot \phi(\text{std}(L))$
Indeed, we extract the first u which corresponds to the shift of the first record.

Central term: $\mu_{p+1}(\pi)$

By definition, $\mu_{p+1}(\pi) = 1 - M_p(\pi)$ if $1 > M_p(\pi)$, and $\mu_{p+1}(\pi) = 0$ otherwise. We distinguish two cases according to whether L is empty or not.

Case $p \geq 1$ (L non-empty). By Lemma 4.1, $\text{Val}(L) = \{2, \dots, p+1\}$, so $M_p(\pi) = \max(\pi_1, \dots, \pi_p) = \max(\text{Val}(L)) = p+1 \geq 2$. Since $1 < p+1 = M_p(\pi)$, the value 1 is not a record: $\mu_{p+1}(\pi) = 0$, and the central block is $u^0d = d$.

Case $p = 0$ (L empty). Then $p+1 = 1$: the value 1 occupies the first position of π . By convention $M_0(\pi) = 0$, so $1 > M_0(\pi)$ and 1 is a record: $\mu_1(\pi) = 1 - 0 = 1$, and the central block is $u^1d = ud$.

In both cases we obtain the same key identity:

$$M_{p+1}(\pi) = M_p(\pi) + \mu_{p+1}(\pi) = p+1$$

For $p \geq 1$: $(p+1) + 0$ and For $p = 0$: $0 + 1$.

This identity is the one invoked below in the computation of the block corresponding to R .

Computation of the block corresponding to R

By Corollary 4.1, $R_j = \text{std}(R)_j + (p+1)$ for all $j \in \{1, \dots, q\}$. We work directly with the running maximum of π itself, rather than with a standalone record-increase sequence of R . Recall from the computation of the central term that $M_{p+1}(\pi) = p+1$ (established above for every $p \geq 0$). For $j \in \{0, \dots, q\}$, set $M'_j := M_{p+1+j}(\pi)$. We show by induction on j that $M'_j = b_j + (p+1)$, where $b_j = M_j(\text{std}(R))$.

Base case ($j = 0$): $M'_0 = M_{p+1}(\pi) = p+1 = b_0 + (p+1)$, since $b_0 = 0$.

Induction step: Assume $M'_{j-1} = b_{j-1} + (p+1)$. The next entry of π is $\pi_{p+1+j} = R_j = \text{std}(R)_j + (p+1)$. We have $\pi_{p+1+j} > M'_{j-1} \iff \text{std}(R)_j + (p+1) > b_{j-1} + (p+1) \iff \text{std}(R)_j > b_{j-1}$, i.e. π_{p+1+j} is a record of π if and only if $\text{std}(R)_j$ is a record of $\text{std}(R)$. At record positions, $\mu_{p+1+j}(\pi) = \pi_{p+1+j} - M'_{j-1} = \text{std}(R)_j - b_{j-1} = \beta_j$, and at non-record positions $\mu_{p+1+j}(\pi) = 0 = \beta_j$. In both cases $\mu_{p+1+j}(\pi) = \beta_j$, and consequently $M'_j = M'_{j-1} + \mu_{p+1+j}(\pi) = (b_{j-1} + (p+1)) + \beta_j = (b_{j-1} + \beta_j) + (p+1) = b_j + (p+1)$, using the telescoping identity $b_j = b_{j-1} + \beta_j$ for $\text{std}(R)$. This completes the induction.

In particular, we have shown $\mu_{p+1+j}(\pi) = \beta_j$ for every $j \in \{1, \dots, q\}$: the record-increase sequence of π , restricted to the positions occupied by R , coincides term by term with the record-increase sequence $(\beta_1, \dots, \beta_q)$ of $\text{std}(R)$. Hence the corresponding block of $\phi(\pi)$ is $u^{\mu_{p+2}(\pi)}d u^{\mu_{p+3}(\pi)}d \cdots u^{\mu_{p+1+q}(\pi)}d = u^{\beta_1}d u^{\beta_2}d \cdots u^{\beta_q}d = \phi(\text{std}(R))$.

Assembly

Recall that $\phi(\pi) = \phi(L) \cdot u^{\mu_{p+1}(\pi)}d \cdot \phi(R)$.

If $p \geq 1$: the paragraph "Computation of $\phi(L)$ " gives $\phi(L) = u \cdot \phi(\text{std}(L))$, and the central block equals d . Hence

$$\phi(\pi) = u \cdot \phi(\text{std}(L)) \cdot d \cdot \phi(R).$$

If $p = 0$: $L = \emptyset$, so $\phi(L) = \varepsilon$ and $\text{std}(L) = \emptyset$, so $\phi(\text{std}(L)) = \varepsilon$. The central block equals ud , hence

$$\phi(\pi) = \varepsilon \cdot ud \cdot \phi(R) = u \cdot \varepsilon \cdot d \cdot \phi(R) = u \cdot \phi(\text{std}(L)) \cdot d \cdot \phi(R).$$

In both cases we obtain $\phi(\pi) = u \cdot \phi(\text{std}(L)) \cdot d \cdot \phi(\text{std}(R))$, which completes the proof. $\square$

Remark 4.1. The paragraph "Computation of the block corresponding to R " relies on the identity $M_{p+1}(\pi) = p+1$ as the base case of its induction on j . This identity is now established above for every $p \geq 0$, so that paragraph requires no further modification. Similarly, the paragraph "Computation of $\phi(L)$ " is only invoked in the case $p \geq 1$, where its implicit assumption (that L is non-empty, so that α_1 exists) is now made explicit.

4.2 Binary tree compatibility

Lemma 4.3. $\gamma(\phi(\pi)) = \text{Node}(\gamma(\phi(\text{std}(L))), \gamma(\phi(\text{std}(R))))$

Proof. Let $A = \phi(\text{std}(L))$ and $B = \phi(\text{std}(R))$. By Proposition 3.2, A and B are Dyck words (possibly empty if L or R is empty). By Lemma 4.2: $\phi(\pi) = u \cdot A \cdot d \cdot B$

To apply the definition of γ , we must show that uAd is the smallest prefix of $\phi(\pi)$ that returns to height 0.

- After the initial u : $h(u) = 1$

- During the reading of A : For every prefix A' of A , because A is a Dyck word, we have $h(A') \geq 0$. The current height in uA is $1 + h(A') \geq 1 > 0$. So the height does not reach 0 while reading of A .

- After the final d of uAd : since A is a Dyck Word, its height is always non-negative and $h(A) = 0$. Thus, after reading uAd , the height is $1 + 0 - 1 = 0$

Therefore, uAd is the smallest prefix of $\phi(\pi)$ that returns to height 0. By definition of the canonical decomposition (Lemma 3.1) and of γ (Definition 3.3): $\gamma(\phi(\pi)) = \text{Node}(\gamma(A), \gamma(B)) = \text{Node}(\gamma(\phi(\text{std}(L))), \gamma(\phi(\text{std}(R))))$ $\square$

Lemma 4.4. For every sequence of distinct integers τ , we have $\beta_{\min}(\tau) = \beta_{\min}(\text{std}(\tau))$

Proof. We proceed by induction on $|\tau|$

Base case: If τ is empty, both sides are the empty tree: $\emptyset$.

Induction step: Let $m = \min(\tau)$ and write $\tau = L \cdot m \cdot R$. Then $\text{std}(\tau) = \text{std}(L) \cdot 1 \cdot \text{std}(R)$ because the minimum of τ becomes 1 in $\text{std}(\tau)$ and the relative orders of L and R are preserved by standardization.

By definition of $\beta_{\min}$ (Definition 3.6): $\beta_{\min}(\tau) = \text{Node}(\beta_{\min}(L), \beta_{\min}(R))$

By the induction hypothesis: $\beta_{\min}(L) = \beta_{\min}(\text{std}(L))$, $\beta_{\min}(R) = \beta_{\min}(\text{std}(R))$

Thus $\beta_{\min}(\tau) = \beta_{\min}(\text{std}(\tau))$ $\square$

4.3 Principal theorem

Theorem 4.1. For every $n \geq 1$ and every permutation $\pi \in Av_n(312)$: $\gamma(\phi(\pi)) = \beta_{\min}(\pi)$

Proof. We proceed by induction on n .

Base case: $n = 1$. $\pi = [1]$, $L = \emptyset$, $R = \emptyset$. We have $\phi([1]) = u^1 d = ud$. The canonical decomposition of ud is $u \cdot (\emptyset) \cdot d \cdot (\emptyset)$, so $\gamma(ud) = \text{Node}(\gamma(\emptyset), \gamma(\emptyset)) = \text{Node}(\emptyset, \emptyset) = \text{leaf}$ (a node whose two subtrees are empty).

Moreover, $\beta_{\min}([1]) = \text{Node}(\beta_{\min}(\emptyset), \beta_{\min}(\emptyset)) = \text{leaf}$.

The equality holds.

Induction step: Suppose that the theorem holds for all 312-avoiding permutations of size less than n . Let $\pi \in Av_n(312)$. After normalization, we write $\pi = L \cdot 1 \cdot R$.

Justification that $\text{std}(L), \text{std}(R) \in Av(312)$: Every subsequence of π is still 312-avoiding (any pattern in L or R would also be a pattern in π). Standardization preserves relative order, so it preserves pattern avoidance. Thus $\text{std}(L), \text{std}(R) \in Av(312)$.

By Lemma 4.3: $\gamma(\phi(\pi)) = \text{Node}(\gamma(\phi(\text{std}(L))), \gamma(\phi(\text{std}(R))))$

Because $|\text{std}(L)| = |L| < n$ and $|\text{std}(R)| = |R| < n$, the induction hypothesis can be applied:
 $\gamma(\phi(\text{std}(L))) = \beta_{\min}(\text{std}(L))$, $\gamma(\phi(\text{std}(R))) = \beta_{\min}(\text{std}(R))$
 By Lemma 4.4, $\beta_{\min}(\text{std}(L)) = \beta_{\min}(L)$ and $\beta_{\min}(\text{std}(R)) = \beta_{\min}(R)$.

Finally: $\gamma(\phi(\pi)) = \text{Node}(\beta_{\min}(L), \beta_{\min}(R)) = \beta_{\min}(\pi)$

□

Corollary 4.2 (Bijectivity of ϕ). *For every $n \geq 1$, the map $\phi : Av_n(312) \rightarrow \mathcal{D}_n$ is a bijection.*

Proof. By Theorem 4.1, $\gamma \circ \phi = \beta_{\min}$. Since $\gamma : \mathcal{D}_n \rightarrow \mathcal{T}_n$ is a bijection (Fact 3.1) and $\beta_{\min} : Av_n(312) \rightarrow \mathcal{T}_n$ is a bijection (Fact 3.2), we deduce $\phi = \gamma^{-1} \circ \beta_{\min}$, which is a composition of two bijections, hence a bijection.

□

5 Inverse Bijection Ψ

We now show that the map ϕ allows an explicit inverse description.

We recall that, by Proposition 3.1, every permutation $\pi \in S_n$ is sent to a sequence $(\mu_1, \dots, \mu_n) \in \mathcal{R}_n$.

Conversely, we will show how to directly reconstruct the permutation from such a sequence.

Definition 5.1 (Map Ψ). *Let $\mu = (\mu_1, \dots, \mu_n) \in \mathcal{R}_n$. For all $i \geq 1$, $M_i = \sum_{k=1}^i \mu_k$ so that M_i represents the current record that the reconstructed permutation must have.*

We recursively construct a sequence $\pi = (\pi_1, \dots, \pi_n)$

We initialise with $\pi_1 = M_1 = \mu_1$ and set $\rho = M_1$ where ρ is the current record.

Suppose now, that the values $\pi_1, \dots, \pi_i$ are already constructed.

- *If $\mu_{i+1} > 0$, set $\pi_{i+1} = \rho + \mu_{i+1} = M_{i+1}$, and update the current record $\rho = \pi_{i+1}$*
- *If $\mu_{i+1} = 0$, set $\pi_{i+1} = \max(\{1, \dots, \rho - 1\} \setminus \{\pi_1, \dots, \pi_i\})$.*

The resulting sequence is denoted $\Psi(\mu)$.

Lemma 5.1. *For every $i \in \{1, \dots, n\}$, after constructing the first i values of $\Psi(\mu)$ the current record $\rho = M_i = \sum_{k=1}^i \mu_k$ and it is the highest value constructed so far.*

Proof. We proceed by induction on i

Base case ($i = 1$): $\pi_1 = \mu_1 = M_1$. The current record is $\rho = M_1$, which is the highest constructed value because it is the only one.

Induction step: Suppose the property is in effect up to step i .

- If $\mu_{i+1} > 0$, then by the definition of the algorithm, $\pi_{i+1} = \rho + \mu_{i+1}$. Since $\mu_{i+1} > 0$, $\pi_{i+1} > \rho$.

Moreover, $\rho + \mu_{i+1} = M_i + \mu_{i+1} = M_{i+1}$

The new current record is M_{i+1} , which is also the highest built value.

- If $\mu_{i+1} = 0$, then the algorithm chooses a value strictly less than ρ . The current record is not updated and remains equal to $M_i = M_{i+1}$. Since the new value is strictly less than ρ , ρ remains the highest built value.

In both cases, the property holds.

□

Lemma 5.2. *The map Ψ is well defined, i.e. when $\mu_{i+1} > 0$, the set $\{x < \rho \mid x \notin \{\pi_1, \dots, \pi_i\}\}$ is not empty.*

Proof. According to Lemma 5.1, the current record is $R = M_i$. Since $\mu \in \mathcal{R}_n$, $M_i = \sum_{k=1}^i \mu_k \geq i$.

The integers strictly less than R are exactly $1, 2, \dots, R-1$ a total of $R-1$.

By Lemma 5.1 again, among the i values already constructed, only one is equal to R , the current record. The remaining values of $i-1$ are strictly less than R .

Thus, at most $i-1$ integers of $\{1, \dots, R-1\}$ have already been used.

But, $R-1 \geq i-1$. so at least one integer strictly less than R is not yet used.

Therefore, the considered set is non-empty, which shows that the construction is well defined. $\square$

Lemma 5.3 ($\Psi(\mu)$ is a permutation). *For every $\mu = (\mu_1, \dots, \mu_n) \in \mathcal{R}_n$, the sequence $\pi = \Psi(\mu)$ is a permutation of $\{1, \dots, n\}$, i.e. $\pi \in S_n$.*

Proof. Injectivity. We show by induction on i that $\pi_1, \dots, \pi_i$ are pairwise distinct.

Base case ($i = 1$): trivial, a single value.

Induction step: Suppose $\pi_1, \dots, \pi_i$ are pairwise distinct, and consider π_{i+1} .

- If $\mu_{i+1} > 0$: $\pi_{i+1} = M_{i+1}$. By Lemma 5.1, M_i is the largest value among $\pi_1, \dots, \pi_i$, so $M_i \geq \pi_j$ for all $j \leq i$. Since $\mu_{i+1} > 0$, we have $M_{i+1} = M_i + \mu_{i+1} > M_i \geq \pi_j$ for all $j \leq i$. Thus π_{i+1} is strictly greater than all previous values, hence distinct from each of them.

- If $\mu_{i+1} = 0$: by definition, $\pi_{i+1} \in \{1, \dots, \rho-1\} \setminus \{\pi_1, \dots, \pi_i\}$, so $\pi_{i+1} \notin \{\pi_1, \dots, \pi_i\}$ explicitly.

In both cases, $\pi_{i+1} \notin \{\pi_1, \dots, \pi_i\}$, so $\pi_1, \dots, \pi_{i+1}$ are pairwise distinct. By induction, $\pi_1, \dots, \pi_n$ are pairwise distinct.

Values in $\{1, \dots, n\}$. For every i , $\pi_i \leq M_i$ (equality if $\mu_i > 0$. Otherwise $\pi_i < \rho = M_{i-1} \leq M_i$ by (C1)). Since (M_i) is nondecreasing (a partial sum of $\mu_k \geq 0$ by (C1)), $M_i \leq M_n$ for every $i \leq n$. By (C2), $M_n = n$. Hence $\pi_i \leq n$ for every i . Moreover $\pi_i \geq 1$ by construction. Thus $\pi_i \in \{1, \dots, n\}$ for every i .

Conclusion. π is an injective map from $\{1, \dots, n\}$ to $\{1, \dots, n\}$: n pairwise distinct values, all in a set of cardinality n . By the pigeonhole principle, π is bijective, hence $\pi \in S_n$. $\square$

Lemma 5.4 (Block structure). *Let $\pi = \Psi(\mu)$. Consider two successive records $R_j < R_{j+1}$ in the construction of π . The values built between these two records form a decreasing block.*

This block is built first with the values strictly between R_j and R_{j+1} , then possibly from the values strictly less than R_j .

Proof. Consider a block associated with the current record R_j . According to the definition of the algorithm, while no new record is created, we have: $\mu_{i+1} = 0$ and therefore: $\pi_{i+1} = \max(\{1, \dots, R_j - 1\} \setminus \{\pi_1, \dots, \pi_i\})$. Thus, at each step, the chosen value is the highest available value still strictly less than the current record.

Therefore, the values obtained in this block are strictly decreasing.

We now describe which values appear in this block. At the moment when the new record R_{j+1} is created, all values between the two records: $R_j < x < R_{j+1}$ become available for the first time, because they were previously greater than the current record.

Since the algorithm always chooses the greatest available value, these new values are necessarily used before the values already available from the previous blocks. After all values in the interval $]R_j, R_{j+1}[$ are used, the algorithm continues with the remaining values, which are all less than R_j .

Therefore, each block is of the form: $\underbrace{a_k > a_{k-1} > \dots > a_1}_{R_j < a_1 < R_{j+1}} > \underbrace{b_l > \dots > b_1}_{b_i < R_j}$

The block is decreasing and satisfies the announced property. $\square$

Lemma 5.5 (Monotonicity below a threshold). *Let $\mu \in \mathcal{R}_n$ and $\pi = \Psi(\mu)$. Fix an index $i_1 \in \{1, \dots, n\}$ and set $v = \pi_{i_1}$. Then the subsequence of terms π_j with $j > i_1$ and $\pi_j < v$ is strictly decreasing.*

Proof. Recall that $\rho_j := M_j$ denotes the current record after j steps of the construction of π , and that $(\rho_j)_{j \geq 0}$ is non-decreasing. Since $v = \pi_{i_1} \leq \max(\pi_1, \dots, \pi_{i_1}) = \rho_{i_1}$, we have $\rho_j \geq \rho_{i_1} \geq v$ for every $j \geq i_1$.

Fix $j \geq i_1$ and, in the case $\mu_{j+1} = 0$, split the set of available values at step $j+1$ as $\text{pool}_j := \{1, \dots, \rho_j - 1\} \setminus \{\pi_1, \dots, \pi_j\} = \underbrace{(\text{pool}_j \cap \{1, \dots, v-1\})}_{\text{low part}} \cup \underbrace{(\text{pool}_j \cap \{v, \dots, \rho_j - 1\})}_{\text{high part}}$.

Step 1: An integer $x < v$ belongs to pool_j as soon as it is *eligible* (i.e. $x < \rho_j$, which holds for every $j \geq i_1$ since $\rho_j \geq v > x$) and unused.

The only mechanism that adds new integers to the pool is a record jump, which introduces the interval $(\rho_{\text{old}}, \rho_{\text{new}})$.

For any such jump occurring at a step $> i_1$, we have $\rho_{\text{old}} \geq \rho_{i_1} \geq v$, so every newly introduced integer is $> v$, never $< v$. Hence, from step i_1 onward, the low part can only lose elements (when one is chosen), never gain any.

Step 2: By definition of Ψ , a non-record step selects $\max(\text{pool}_j)$.

If the high part is non-empty, every element of the high part is $\geq v$. Hence strictly greater than every element of the low part. Thus the maximum is taken in the high part, and the chosen value is $\geq v$. In particular it is not one of the terms we are tracking.

Step 3: conclusion. A value $\pi_j < v$ can only be produced at a step where the high part is empty, in which case

$$\pi_j = \max(\text{low part at that time})$$

By Step 1, the low part is non-increasing (as a set, under inclusion) from i_1 onward.

So if $\pi_{j_2} < v$ is chosen at step j_2 and $\pi_{j_3} < v$ is chosen at a later step $j_3 > j_2$, then π_{j_3} is the maximum of a subset of the low part from which π_{j_2} has been removed. Hence $\pi_{j_3} < \pi_{j_2}$.

This shows that the subsequence of terms π_j ($j > i_1$) with $\pi_j < v$ is strictly decreasing. $\square$

Corollary 5.1 ($\Psi(\mu)$ avoids 312). *For every $\mu \in \mathcal{R}_n$, the sequence $\pi = \Psi(\mu)$ avoids the pattern 312. In particular $\Psi(\mathcal{R}_n) \subseteq \text{Av}_n(312)$.*

Proof. Suppose, for contradiction, that there exist indices $i_1 < i_2 < i_3$ with $\pi_{i_2} < \pi_{i_3} < \pi_{i_1}$.

Set $v = \pi_{i_1}$. Then both π_{i_2} and π_{i_3} are terms with index $> i_1$ and value $< v$, occurring at positions $i_2 < i_3$ with $\pi_{i_2} < \pi_{i_3}$. This contradicts Lemma 5.5, which forces the subsequence of such terms to be strictly decreasing, i.e. $\pi_{i_2} > \pi_{i_3}$.

Hence no such triple of indices exists, and $\pi = \Psi(\mu)$ avoids 312. $\square$

Proposition 5.1. *For every sequence $\mu = (\mu_1, \dots, \mu_n) \in \mathcal{R}_n$ If $\pi = \Psi(\mu)$, then for every i $\mu_i(\pi) = \mu_i$ In particular, $\phi(\Psi(\mu)) = u^{\mu_1} \cdot d \cdot u^{\mu_2} \cdot d \dots u^{\mu_n} \cdot d$*

Proof. By Lemma 5.3, $\pi = \Psi(\mu) \in S_n$, so $\phi(\pi)$ is well defined.

Let $\mu = (\mu_1, \dots, \mu_n) \in \mathcal{R}_n$ and set $\pi = \Psi(\mu)$.

For every i , denote $M_i = \sum_{k=1}^i \mu_k$.

By Lemma 5.1, after constructing the first i values of π , the current record is exactly M_i .

Consider now an index i .

- If $\mu_i > 0$, then, according to the definition of Ψ we have $\pi_i = M_i = M_{i-1} + \mu_i$.

Since $M_i > M_{i-1}$, the value π_i is a new record of π . According to the definition of ϕ (Definition 3.10), $\mu_i(\pi) = \pi_i - M_{i-1} = M_i - M_{i-1} = \mu_i$

- If $\mu_i = 0$, then the algorithm chooses a value strictly less than the current record M_{i-1} . Thus, $\pi_i < M_{i-1}$ and π_i is not a record.

According to the definition of ϕ , $\mu_i(\pi) = 0 = \mu_i$.

In both cases, $\mu_i(\pi) = \mu_i$.

The record-increase sequences of π and of μ coincide term by term, which shows $\phi(\Psi(\mu)) = u^{\mu_1} d \dots u^{\mu_n} d$ $\square$

Lemma 5.6 ($\mathcal{R}_n \leftrightarrow \mathcal{D}_n$). *The map $\Phi_0 : \mathcal{R}_n \rightarrow \mathcal{D}_n$ defined by $\Phi_0(\mu) = u^{\mu_1} d u^{\mu_2} d \dots u^{\mu_n} d$ is a bijection.*

Proof. Φ_0 **is well defined.** Let $\mu = (\mu_1, \dots, \mu_n) \in \mathcal{R}_n$. The same argument as in Proposition 3.2 (which only used properties (C1)-(C3) of μ , not that μ comes from a permutation) shows that $\Phi_0(\mu)$ satisfies (D1) and (D2), hence $\Phi_0(\mu) \in \mathcal{D}_n$.

Injectivity. Let $\mu, \mu' \in \mathcal{R}_n$ with $\Phi_0(\mu) = \Phi_0(\mu')$. The word $w = \Phi_0(\mu)$ has exactly n occurrences of the letter d , at positions $\mu_1 + 1, (\mu_1 + \mu_2 + 2), \dots$

More precisely, the i -th letter d of w is immediately preceded by a run of μ_i consecutive letters u which is possibly empty. Itself immediately preceded by the $(i-1)$ -th letter d or by the start of the word if $i = 1$.

Thus μ_i is recovered from w as the number of u 's between the $(i-1)$ -th and i -th occurrences of d . Since $\Phi_0(\mu) = \Phi_0(\mu')$ as words, this recovery gives $\mu_i = \mu'_i$ for every i , so $\mu = \mu'$.

Surjectivity. Let $w \in \mathcal{D}_n$. By (D1), w contains exactly n letters d . Define μ_i as the number of letters u appearing between the $(i-1)$ -th and i -th occurrence of d in w . This is with the convention that the 0-th occurrence marks the start of w .

By construction,

$$w = u^{\mu_1} d u^{\mu_2} d \dots u^{\mu_n} d = \Phi_0(\mu)$$

and $\mu_i \geq 0$ for all i , so (C1) holds.

Since w has length $2n$ and contains n letters d , it also contains n letters u , so $\sum_{i=1}^n \mu_i = n$, which is (C2).

Finally, for $k \in \{1, \dots, n\}$, the prefix of w ending at the k -th letter d (inclusive) contains $\sum_{i=1}^k \mu_i$ letters u and exactly k letters d . By (D2) its height is non-negative, i.e. $\sum_{i=1}^k \mu_i - k \geq 0$, which is (C3). Hence $\mu \in \mathcal{R}_n$, and $\Phi_0(\mu) = w$. $\square$

Remark 5.1. *Throughout this section, we freely identify $\phi(\pi) \in \mathcal{D}_n$ (as in Proposition 3.2) with its record-increase sequence $(\mu_1(\pi), \dots, \mu_n(\pi)) \in \mathcal{R}_n$, via the bijection Φ_0 of Lemma 5.6. In particular, Ψ is defined equivalently as a map on $\mathcal{R}_n$ or on $\mathcal{D}_n$.*

Corollary 5.2 (Ψ is the inverse of ϕ). *The map Ψ is the inverse of ϕ .*

Proof. By Corollary 5.1, $\Psi(\mathcal{R}_n) \subseteq Av_n(312)$, independently of any bijectivity assumption on ϕ .

By Corollary 4.2 (following Theorem 4.1) together with the $\mathcal{R}_n \leftrightarrow \mathcal{D}_n$ identification of Lemma 5.6, the map $\phi : Av_n(312) \rightarrow \mathcal{R}_n$ is a bijection. In particular $|\mathcal{R}_n| = |Av_n(312)| = C_n$.

Proposition 5.1 shows that $\phi \circ \Psi = \text{Id}_{\mathcal{R}_n}$. Combined with $\Psi(\mathcal{R}_n) \subseteq Av_n(312)$, this means Ψ is injective on $\mathcal{R}_n$ (a left inverse of an injective map is itself injective on the relevant domain) and its image lies in $Av_n(312)$. Since ϕ restricted to $Av_n(312)$ is a bijection onto $\mathcal{R}_n$ and $\phi \circ \Psi = \text{Id}$, Ψ must coincide with the (two-sided) inverse of this restriction on all of $\mathcal{R}_n$.

Therefore $\Psi = \phi^{-1}$. □

6 Lattice isomorphism

Theorem 6.1. *For every $n \geq 1$, the map ϕ is a lattice isomorphism $\phi : (Av_n(312), \leq_W) \rightarrow (\mathcal{D}_n, \leq_{\text{Tamari}})$*

Proof. Step 1: ϕ is a bijection

By Corollary 4.2, ϕ is a bijection.

Step 2: ϕ preserves the order in both directions.

Let $\pi, \sigma \in Av_n(312)$. We have: $\pi \leq_W \sigma$
 $\iff \text{Inv}(\pi) \subseteq \text{Inv}(\sigma)$ (definition of $\leq_W$ on $Av_n(312)$, Def. 3.7)
 $\iff \beta_{\min}(\pi) \leq_{\text{Tamari}} \beta_{\min}(\sigma)$ (Fact 3.2: $\beta_{\min}$ is a lattice isomorphism)
 $\iff \gamma(\phi(\pi)) \leq_{\text{Tamari}} \gamma(\phi(\sigma))$ (Theorem 4.1: $\gamma \circ \phi = \beta_{\min}$)
 $\iff \phi(\pi) \leq_{\text{Tamari}} \phi(\sigma)$ (Fact 3.1: γ is a lattice isomorphism)

Step 3: ϕ is a lattice isomorphism

ϕ is a bijection between two finite lattices that preserves the order in both directions. In a finite lattice, such a bijection necessarily preserves infima and suprema: for every $\pi, \sigma \in Av_n(312)$, $\phi(\pi \wedge \sigma) = \phi(\pi) \wedge \phi(\sigma)$, $\phi(\pi \vee \sigma) = \phi(\pi) \vee \phi(\sigma)$

Therefore $x \leq y \iff x \wedge y = x$ in any lattice, ϕ is a lattice isomorphism. □

We have constructed an explicit bijection ϕ between the 312-avoiding permutations and Dyck paths. This bijection is in fact a lattice isomorphism, which shows that the right weak order restricted to $Av_n(312)$ coincides with the Tamari order on the Dyck words.

7 Transported statistics by ϕ

Proposition 7.1 (Records $\longleftrightarrow$ Peaks). *Let $\pi \in Av_n(312)$. Let $\text{rec}(\pi)$ the number of records of π , i.e. the number of indices i such that $\pi_i > M_{i-1}$. Let $\text{peak}(w)$ the number of peaks of a Dyck word w , i.e. the number of factors ud in w . Then $\text{rec}(\pi) = \text{peak}(\phi(\pi))$*

Proof. By definition $\phi(\pi) = u^{\mu_1} d u^{\mu_2} d \dots u^{\mu_n} d$

By the definition of record increase (Definition 3.9), $\mu_i > 0$ if and only if π_i is a record.

Consider an index i

If $\mu_i > 0$, the corresponding block is $u^{\mu_i} d$

The last u of this block is immediately followed by the d , so exactly one ud factor appears, i.e. exactly one peak.

If $\mu_i = 0$, the block reduces to a single d and contains no ud factor.

Thus each i with $\mu_i > 0$ contributes exactly one peak and no other peaks appear.

Therefore $\text{peak}(\phi(\pi)) = |\{i : \mu_i > 0\}| = \text{rec}(\pi)$ □

Proposition 7.2 (Narayana records distribution). *For every integer $n \geq 1$ and $1 \leq k \leq n$ $|\{\pi \in \text{Av}_n(312) : \text{rec}(\pi) = k\}| = N(n, k)$ where $N(n, k) = \frac{1}{n} \binom{n}{k} \binom{n}{k-1}$ is the number of Narayana (The sum of the Narayana number in the n -th line of the Narayana triangle is equal to the n -th Catalan number. We have $\sum_{k=1}^n N(n, k) = C_n$).*

Proof. The map $\phi : \text{Av}_n(312) \rightarrow \mathcal{D}_n$ is a bijection between the 312-avoiding permutations and Dyck paths of semilength n . By the previous Proposition 7.1, $\text{rec}(\pi) = \text{peak}(\phi(\pi))$. Thus ϕ preserves exactly the number of records and the number of peaks.

But the Narayana number $N(n, k)$ counts the Dyck paths of semilength n with exactly k peaks.

Since ϕ is a bijection, the number of 312-avoiding permutations with exactly k records is also $N(n, k)$.

Hence $|\{\pi \in \text{Av}_n(312) : \text{rec}(\pi) = k\}| = N(n, k)$ □

Proposition 7.3 (Height at block endpoints). *Let $\pi \in \text{Av}_n(312)$. After reading the first k blocks $u^{\mu_1}d \dots u^{\mu_k}d$, the height of the path $\phi(\pi)$ is $M_k - k$.*

Proof. Let $\pi \in \text{Av}_n(312)$ and $M_k = \max(\pi_1, \dots, \pi_k)$. By the telescoping Lemma 3.2, $M_k = \sum_{i=1}^k \mu_i$. After reading the first k blocks $u^{\mu_1}d \dots u^{\mu_k}d$ The height of the path is $\sum_{i=1}^k (\mu_i - 1) = \left(\sum_{i=1}^k \mu_i\right) - k = M_k - k$. Thus the heights reached at the block endpoints are exactly $M_k - k$. The maximal height at a block endpoint is $\max_{1 \leq k \leq n} (M_k - k)$. The vertical geometry of $\phi(\pi)$ is entirely determined by the record sequence. □

Remark 7.1. *The quantity $M_k - k$ measures the gap between the largest value seen so far and the number of positions already read.*

When records appear early, M_k grows quickly and the path rises high. When records are delayed, $M_k - k$ stays small and the path remains close to the axis.

Thus, the vertical geometry of the path $\phi(\pi)$ directly reflects the growth rate of records of π . In particular, the current maxima sequence $(M_1, \dots, M_n)$ completely determines the height of the path at block endpoints.

Proposition 7.4 (Maximal height). *Let $\pi \in \text{Av}_n(312)$. The maximal height reached by the Dyck path $\phi(\pi)$ is $H(\phi(\pi)) = \max_{1 \leq k \leq n} (M_k - k) + 1$*

Proof. Write $\phi(\pi) = u^{\mu_1}d u^{\mu_2}d \dots u^{\mu_n}d$. After reading the first $k-1$ blocks, the height of the path is $M_{k-1} - (k-1)$. The k -th block is of the form $u^{\mu_k}d$. During the μ_k climbs, the height increases at each letter u of one unit. Thus the maximal height in this block is $M_{k-1} - (k-1) + \mu_k$. But, by definition of record increase, $\mu_k = M_k - M_{k-1}$. We have $M_{k-1} - (k-1) + \mu_k = M_k - k + 1$. Since the block consists of consecutive up-steps followed by one down-step, the height is strictly increasing throughout the ascent and decreases only at the final descent. Thus, $H(\phi(\pi)) = \max_{1 \leq k \leq n} (M_k - k) + 1$ □

Definition 7.1 (Area). *Let $w = u^{\mu_1}du^{\mu_2}d \cdots u^{\mu_n}d$ a Dyck word.*

Its area (in this article) is defined as the sum of the heights at which the descents start, i.e.

$$\text{Area}(w) = \sum_{k=1}^n h_k$$

where h_k denotes its height just before the k -th letter d .

Proposition 7.5 (Area). *Let $\pi \in Av_n(312)$. Then*

$$\text{Area}(\phi(\pi)) = \sum_{k=1}^n (M_k - k + 1)$$

Proof. Write

$$\phi(\pi) = u^{\mu_1}du^{\mu_2}d \cdots u^{\mu_n}d$$

By Proposition 7.3, after reading the first $k - 1$ blocks, the height of the path is

$$M_{k-1} - (k - 1)$$

The k -th block is of the form

$$u^{\mu_k}d$$

Just before the final descent of the block, the height is

$$M_{k-1} - (k - 1) + \mu_k$$

Since,

$$\mu_k = M_k - M_{k-1}$$

this equals

$$M_{k-1} - (k - 1) + \mu_k = M_k - k + 1$$

Summing over all k gives:

$$\text{Area}(\phi(\pi)) = \sum_{k=1}^n (M_k - k + 1)$$

□

Unlike the original Bandlow–Killpatrick construction, the record-increment formulation makes these statistics immediate to compute from the current maxima sequence. No recursive decomposition or adjacent-transposition process is required.

The bijection ϕ encodes not only the recursive structure of the 312-avoiding permutations, but also several of their statistics in a simple way.

8 Link between [BK01] and our bijection

We now make precise in what sense ϕ coincides with the bijection of Bandlow and Killpatrick [BK01]. We first recall their construction in a convenient equivalent form. This is obtained by unwinding the successive adjacent transpositions of the original definition into a single insertion procedure.

Definition 8.1 (Insertion). *Given a sequence $\sigma = (\sigma_1, \dots, \sigma_r)$ of distinct integers, an integer $v \notin \{\sigma_1, \dots, \sigma_r\}$, and a position q with $0 \leq q \leq r$. We denote by $\text{ins}(\sigma, v, q)$ the sequence of length $r + 1$ obtained by inserting v immediately after the first q terms of σ :*

$$\text{ins}(\sigma, v, q) = (\sigma_1, \dots, \sigma_q, v, \sigma_{q+1}, \dots, \sigma_r).$$

Definition 8.2 (Profile of a Dyck word). Let $w \in \mathcal{D}_n$. For $j = 0, \dots, n-1$, let $a_j(w)$ denote the height of w immediately before its $(j+1)$ -th letter u . The sequence $(a_0(w), \dots, a_{n-1}(w))$ is called the profile of w .

Definition 8.3 (Bandlow-Killpatrick bijection). Let $w \in \mathcal{D}_n$ with profile $(a_0, \dots, a_{n-1})$. Define a sequence of permutations $\sigma^{(0)} = \emptyset, \sigma^{(1)}, \dots, \sigma^{(n)}$ by

$$\sigma^{(m)} = \text{ins}(\sigma^{(m-1)}, m, (m-1) - a_{m-1}), \quad m = 1, \dots, n.$$

We set $BK(w) := \sigma^{(n)}$.

Lemma 8.1 (Adjacent-transposition chain as insertion). Let $\sigma = (\sigma_1, \dots, \sigma_r)$ be a sequence, let $v \notin \{\sigma_1, \dots, \sigma_r\}$, and let $0 \leq a \leq r$. Define $\tau^{(0)} = (\sigma_1, \dots, \sigma_r, v)$, and for $t = 1, \dots, a$, let $\tau^{(t)}$ be obtained from $\tau^{(t-1)}$ by swapping v with its immediate left neighbor. A single adjacent transposition applied only to v . Then

$$\tau^{(a)} = \text{ins}(\sigma, v, r - a).$$

Proof. We show by induction on t that

$$\tau^{(t)} = (\sigma_1, \dots, \sigma_{r-t}, v, \sigma_{r-t+1}, \dots, \sigma_r).$$

Base case ($t = 0$): $\tau^{(0)} = (\sigma_1, \dots, \sigma_r, v)$, by definition of the formula for $t = 0$.

Induction step: Assume the property holds at rank $t-1$ (with $t \leq a \leq r$):

$$\tau^{(t-1)} = (\sigma_1, \dots, \sigma_{r-t+1}, v, \sigma_{r-t+2}, \dots, \sigma_r).$$

The immediate left neighbor of v in this sequence is σ_{r-t+1} . Swapping v with this neighbor modifies only these two entries.

Hence

$$\tau^{(t)} = (\sigma_1, \dots, \sigma_{r-t}, v, \sigma_{r-t+1}, \dots, \sigma_r)$$

which is the property at rank t .

In particular, at $t = a$:

$$\tau^{(a)} = (\sigma_1, \dots, \sigma_{r-a}, v, \sigma_{r-a+1}, \dots, \sigma_r) = \text{ins}(\sigma, v, r - a). \quad \square$$

Lemma 8.2 (Profile bound). Let $w \in \mathcal{D}_n$ with profile $(a_0(w), \dots, a_{n-1}(w))$. Then

$$a_j(w) \leq j \quad \text{for every } j \in \{0, \dots, n-1\}.$$

Proof. Fix $j \in \{0, \dots, n-1\}$, and consider the prefix w' of w consisting of all letters read before the $(j+1)$ -th occurrence of u . By definition of the profile, $a_j(w) = h(w')$.

Since w' stops immediately before the $(j+1)$ -th letter u of w , it contains exactly j letters u (namely the first j occurrences), together with some number $k \geq 0$ of letters d . Hence

$$a_j(w) = h(w') = |u(w')| - |d(w')| = j - k \leq j,$$

since $k \geq 0$. $\square$

Remark 8.1 (Equivalence with the original construction). *In the original construction of [BK01], at step m (for $m = 1, \dots, n$), one starts from the partial permutation $\sigma^{(m-1)}$ on $\{1, \dots, m-1\}$ (of length $m-1$), places the value m at the end (position m), and then moves it leftward by a_{m-1} successive adjacent transpositions. This is exactly the process $\tau^{(0)} \rightarrow \tau^{(1)} \rightarrow \dots \rightarrow \tau^{(a_{m-1})}$ of the previous lemma, with $r = m-1$, $v = m$, $a = a_{m-1}$.*

By the lemma (with $r - a = (m-1) - a_{m-1}$),

$$\sigma^{(m)} = \text{ins}(\sigma^{(m-1)}, m, (m-1) - a_{m-1}),$$

which is exactly the recursive formula in Definition 8.3. By the Profile bound lemma (Lemma 8.2), $a_{m-1} \leq m-1$, so the insertion position $(m-1) - a_{m-1} \geq 0$ is always valid. This shows that the insertion-based definition we use is a faithful, unwound reformulation of the original column-by-column, adjacent-transposition construction of [BK01].

Lemma 8.3 (Outer-arch recursion for BK). *Let $A \in \mathcal{D}_k$, $B \in \mathcal{D}_m$, and let $w = uAdB \in \mathcal{D}_{k+m+1}$ be the corresponding canonical decomposition. Then*

$$BK(w) = (BK(A) + 1) \cdot (1) \cdot (BK(B) + k + 1),$$

where $\sigma + c$ denotes the entrywise addition of the constant c to a sequence σ .

Proof. Write $\alpha = (a_0, \dots, a_{k-1})$ for the profile of A and $\beta = (b_0, \dots, b_{m-1})$ for the profile of B . Tracing heights through $w = uAdB$ shows that the profile of w is

$$(a_0(w), \dots, a_{n-1}(w)) = (0, 1 + a_0, \dots, 1 + a_{k-1}, b_0, \dots, b_{m-1}).$$

Steps $m = 1, \dots, k+1$. We show by induction on m that

$$\sigma^{(m)} = (BK(A_{\leq m-1}) + 1) \cdot (1)$$

where $A_{\leq j}$ denotes the word obtained after j insertion steps of A 's own construction, and that the value 1 always occupies the last position of $\sigma^{(m)}$.

For $m = 1$: $a_0(w) = 0$, so $\sigma^{(1)} = \text{ins}(\emptyset, 1, 0) = (1)$, the claim holds.

Assume it holds at some m with $1 \leq m \leq k$. Step $m+1$ inserts value $m+2$ at position

$$(m+1-1) - a_m(w) = m - (1 + a_{m-1}) = (m-1) - a_{m-1}$$

which is exactly the position used at the corresponding step of A 's own construction, applied to values shifted by +1.

Since this position is $\leq m-1$, it lies at or before the last slot, occupied by 1, so 1 is pushed one position further right. The insertion among the first $m-1$ shifted entries reproduces $BK(A_{\leq m}) + 1$ exactly. This proves the induction, at $m = k+1$ we get $\sigma^{(k+1)} = (BK(A) + 1) \cdot (1)$.

Steps $m = k+2, \dots, n$. Write $i = m - (k+2)$, so the profile entry used is $a_{k+1+i}(w) = b_i$, and the insertion position is $(m-1) - a_{m-1}(w) = (k+1+i) - b_i \geq k+1$, since $b_i \leq i$ (Lemma 8.2 applied to $B \in \mathcal{D}_m$).

Hence every such insertion occurs at or after position $k+1$: the prefix $(BK(A) + 1) \cdot (1)$ built is never modified again. By re-indexing positions $\geq k+1$ from 0, the insertion position becomes exactly $i - b_i$ which is matching B 's own construction step by step.

By the same induction as above, applied to B , this reproduces $BK(B) + (k+1)$ in the remaining slots. Concatenating both parts gives

$$BK(w) = (BK(A) + 1) \cdot (1) \cdot (BK(B) + k + 1)$$

□

Corollary 8.1 (Outer-arch recursion for Ψ). *With the same notation, $\Psi(w) = (\Psi(A) + 1) \cdot (1) \cdot (\Psi(B) + k + 1)$.*

Proof. Let $\pi = \Psi(w)$ and write its decomposition around the minimum as

$$\pi = L \cdot 1 \cdot R,$$

where $p = |L|$ and $q = |R|$.

By Lemma 4.2,

$$\phi(\pi) = u \cdot \phi(\text{std}(L)) \cdot d \cdot \phi(\text{std}(R)).$$

Since $\phi(\pi) = w$, and the canonical decomposition of w is

$$w = uAdB,$$

Lemma 3.1 implies that this decomposition is unique. Therefore,

$$|A| = 2p, \quad |B| = 2q,$$

so that $p = k$ and $q = m$, and moreover

$$A = \phi(\text{std}(L)), \quad B = \phi(\text{std}(R)).$$

Because $\Psi = \phi^{-1}$ (Corollary 5.2), it follows that

$$\text{std}(L) = \Psi(A), \quad \text{std}(R) = \Psi(B).$$

Finally, Corollary 4.1 gives

$$L = \text{std}(L) + 1 = \Psi(A) + 1,$$

and

$$R = \text{std}(R) + (p + 1) = \Psi(B) + (k + 1),$$

since $p = k$.

Hence,

$$\Psi(w) = \pi = (\Psi(A) + 1) \cdot 1 \cdot (\Psi(B) + k + 1).$$

□

Proposition 8.1 (Equivalence with the Bandlow–Killpatrick bijection). *Our record-increment construction is not a new Catalan correspondence, but an explicit one-pass realization of the inverse Bandlow–Killpatrick bijection.*

Proof. Both Ψ and BK are maps

$$\mathcal{D} \longrightarrow Av(312)$$

satisfying the same initial condition

$$\Psi(\emptyset) = BK(\emptyset) = \emptyset.$$

Moreover, by the previous lemma (Lemma 8.3) and corollary (Corollary 8.1), both maps satisfy the same recursive rule with respect to the canonical decomposition

$$w = uAdB.$$

Indeed,

$$BK(w) = (BK(A) + 1) \cdot 1 \cdot (BK(B) + k + 1),$$

and

$$\Psi(w) = (\Psi(A) + 1) \cdot 1 \cdot (\Psi(B) + k + 1).$$

Since the canonical decomposition is unique (Lemma 3.1), both subwords A and B have strictly smaller semilength than w .

Therefore, the value of either map on w is completely determined by its values on strictly smaller Dyck words.

We now argue by induction on the semilength.

The claim is true for the empty Dyck word. Assume that

$$\Psi(A) = BK(A), \quad \Psi(B) = BK(B)$$

for all Dyck words of semilength smaller than n .

Applying the common recursive rule yields

$$\Psi(w) = (\Psi(A) + 1) \cdot 1 \cdot (\Psi(B) + k + 1) = (BK(A) + 1) \cdot 1 \cdot (BK(B) + k + 1) = BK(w).$$

Hence $\Psi = BK$ on $\mathcal{D}_n$ for every n .

Since $\Psi = \phi^{-1}$, we conclude that

$$\phi = BK^{-1}.$$

□

9 Perspectives and conclusion

The results presented in this article show that the map ϕ provides an explicit and linear bijection between 312-avoiding permutations and Dyck paths which is a reformulation of [BK01].

This construction relies solely on record increments, making the bijection simple and algorithmically efficient: the map runs in $O(n)$ time and is straightforward to implement.

Beyond its explicit nature, ϕ reveals a structural correspondence. We proved that this bijection is a lattice isomorphism between the right weak order restricted to $Av_n(312)$ and the Tamari order on Dyck paths (Theorem 6.1).

The bijection also transports several natural statistics. Records correspond to peaks (Proposition 7.1) and geometric quantities such as height and area translate directly into the current maxima sequence (M_k) (Propositions 7.4 and 7.5). Thus, ϕ not only encodes the structure of 312-avoiding permutations, but also preserves their combinatorial structure.

These observations open up several perspectives. One may investigate additional statistics or extend the construction to other pattern-avoiding classes. The simplicity of ϕ suggests that the increments in the records may serve as unifying principle for describing bijections between Catalan families.

References

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10 Annexe

For experimental tests, we have tested in Python/Sage several lemmas, propositions and theorems in this document. Here the link to the code in github and the outputs.

https://nnicorosi.github.io/page/assets/code_bijection.html

tests of permutations to $n = 11$

1. Test of corollary: check that $L = \{2, 3, \dots, |L| + 1\}$ (Corollary 4.1)

No counterexample found (tests: 82499 permutations)

2. Test of the mu structure: check the recursivity of μ (Lemma 4.2)

always True

3. Test the intermediate conjecture: check the decomposition of ϕ with the minimum

success: 82499/82499 tests pass

4. Test the formula WL: check the direct construction of the left part of ϕ

success: 82499/82499 tests pass

5. Test the formula L: check that $WL = u + \Phi(\text{std}(L))$

success: 82499/82499 tests pass

6. Test of the principal theorem (Theorem 4.1): $\gamma(\Phi(\pi)) = \beta_{\min}(\pi)$

always True